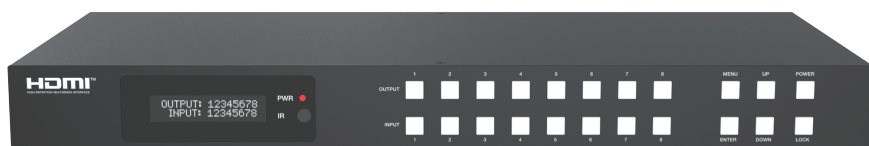


4K30 HDMI™ 8x8 Seamless Matrix



User Manual

VER 1.0

Thank you for purchasing this product

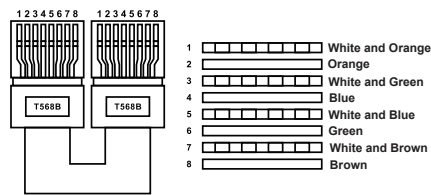
For optimum performance and safety, please read these instructions carefully before connecting, operating or adjusting this product. Please keep this manual for future reference.

Surge protection device recommended

This product contains sensitive electrical components that may be damaged by electrical spikes, surges, electric shock, lighting strikes, etc. Use of surge protection systems is highly recommended in order to protect and extend the life of your equipment.

Caution

The product requires the use of UTP connectors. Please connect in direct interconnection method and do not cross connect.



Direct Interconnection Method

Table of Contents

1. Introduction.....	1
2. Features.....	1
3. Package Contents.....	1
4. Specifications.....	2
5. Operation Controls and Functions.....	3
5.1 Front Panel.....	3
5.2 Rear Panel.....	5
6. IR Remote.....	5
7. LCD Screen Menu Navigation.....	6
8. EDID Management.....	7
9. Web GUI User Guide.....	9
9.1 Connection.....	9
9.2 Login.....	9
9.3 Web GUI Interface Instruction.....	10
10. RS-232 Control Commands.....	15
11. Application Example.....	21

1. Introduction

This 4K30 HDMI Seamless Matrix is a perfect solution for video transmission from 8 HDMI sources to 8 HDTV displays. It supports seamless switching and video wall functionality. 8 channel inputs support video resolution up to 4K30Hz. 8 channel outputs support independent resolution modification, HDMI and DVI output switching, brightness and contrast adjustment. This matrix supports scene preset, saving, calling, deleting and polling. Advanced EDID management is supported. This matrix can be controlled through the front panel buttons, IR remote control, RS-232 commands and Web GUI.

2. Features

- ☆ Supports 10.2Gbps video bandwidth
- ☆ Features 2 operational modes:
 - 8×8 Matrix (seamless switching)
 - Video wall (4×2, 2×2 or 1×8 etc configuration)
- ☆ 8 channel inputs support video resolution up to 4K30Hz, as specified in HDMI 1.4b
- ☆ 8 channel outputs support independent resolution modification:
1024x768@60Hz, 1280x720@50Hz, 1280x720@60Hz, 1280x1024@60Hz,
1360x768@60Hz, 1440x900@60Hz, 1600x900@60Hz, 1600x1200@60Hz,
1920x1080@50Hz, 1920x1080@60Hz, 1920x1200@60Hz, 3840x2160@30Hz,
4096x2160@30Hz
- ☆ 8 channel outputs support HDMI and DVI output switching
- ☆ 8 channel outputs support brightness and contrast adjustment
- ☆ Advanced EDID management, supporting preset EDID, copy EDID, custom EDID (upload)
- ☆ Up to 16 preset scenes, supporting preset scene save, call, delete, polling (the polling time can be set)
- ☆ Control via the front panel buttons, IR remote control, RS-232 commands and Web GUI

3. Package Contents

- ① 1x 4K30 HDMI 8x8 Seamless Matrix
- ② 1x IR Remote
- ③ 2x Mounting Ear
- ④ 8x Machine Screw (KM3*6)
- ⑤ 1x 12V/2.5A Multinational Power Supply (Locking)
- ⑥ 1x User Manual

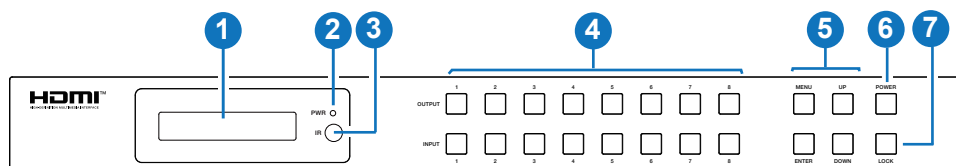
4. Specifications

Technical	
HDMI Compliance	HDMI 1.4b
HDCP Compliance	HDCP 1.4
Video Bandwidth	10.2Gbps
Input/Output Video Resolution	640x480p60Hz, 800x600p60Hz, 1024x768p60Hz, 1280x1024p60Hz, 1360x768p60Hz, 1440x900p60Hz, 1440x1050p60Hz, 1600x1200p60Hz, 1920x1200p60Hz (RB), 720x480i59.94Hz(480i59), 720x480p59.94Hz(480p59), 720x576i50Hz(576i50), 720x576p50Hz(576p50), 1280x720p50Hz(720p50), 1280x720p59.94Hz(720p59), 1280x720p60Hz(720p60), 1920x1080i50Hz(1080i50), 1920x1080i59.94Hz(1080i59), 1920x1080i60Hz(1080i60), 1920x1080p23.98Hz(1080p23), 1920x1080p24Hz(1080p24), 1920x1080p25Hz(1080p25), 1920x1080p29.97Hz(1080p29), 1920x1080p30Hz(1080p30), 1920x1080p50Hz(1080p50), 1920x1080p59.94Hz(1080p59), 1920x1080p60Hz(1080p60), 3840x2160p23.98Hz(2160p23), 3840x2160p24Hz(2160p24), 3840x2160p25Hz(2160p25), 3840x2160p29.97Hz(2160p29), 3840x2160p30Hz(2160p30), 4096x2160p23.98Hz, 4096x2160p24Hz, 4096x2160p25Hz, 4096x2160p29.97Hz, 4096x2160p30Hz
HDR	Support
Color Space	Input: RGB, YCbCr_4:4:4, YCbCr_4:2:2, YCbCr_4:2:0 Output: RGB, YCbCr_4:4:4, YCbCr_4:2:2
Color Depth	8/10/12bit
IR Level	5Vp-p
IR Frequency	Fixed frequency 38KHz
Audio Formats	HDMI input/output: LPCM 2.0/5.1/7.1, Dolby 2.0/5.1, Dolby Digital Plus 2.0/5.1/7.1, DTS 2.0/5.1, DTS HD HRA 5.1/7.1 Analog audio output: PCM 2.0
ESD Protection	IEC 61000-4-2: ±8kV (Air-gap discharge) & ±4kV (Contact discharge)
Connection	
Input	8x INPUT [HDMI Type A, 19-pin female]
Output	8x OUTPUT [HDMI Type A, 19-pin female] 1x AUDIO OUT [3.5mm Stereo Mini-jack]
Control	1x RS-232 [D-Sub 9, female] 1x TCP/IP [RJ45]

Mechanical		
Housing	Metal Enclosure	
Color	Black	
Dimensions	440mm (W) × 203mm (D) × 44.5mm (H)	
Weight	2.56kg	
Power Supply	Input: AC 100~240V 50/60Hz Output: DC 12V/2.5A (US/EU standard, CE/FCC/UL certified)	
Power Consumption	17.52W (MAX)	
Operating Temperature	0°C ~ 40°C / 32°F ~ 104°F	
Storage Temperature	-20°C ~ 60°C / -4°F ~ 140°F	
Operating Humidity	20%~80% (Relative Humidity, non-condensing)	
Storage Humidity	10%~90% (Relative Humidity, non-condensing)	
Recommended HDMI Cable		
Video Resolution	4K30	1080P60
HDMI Cable Length (HDMI IN / OUT)	10m/33ft	15m/49ft
The use of “Premium High Speed HDMI” cable is highly recommended.		

5. Operation Controls and Functions

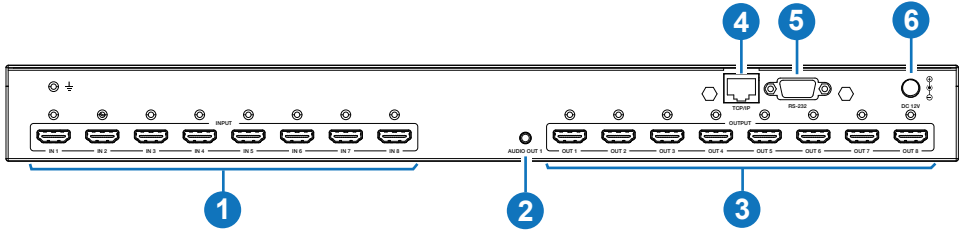
5.1 Front Panel



No.	Name	Function Description
1	LCD screen	The current input/output channel mapping of the matrix is displayed by default. Information such as output resolution, EDID, baud rate, and IP address can be displayed by pressing menu buttons.
2	PWR LED	The LED will light in green when the matrix is working, and red when the matrix is on standby.
3	IR window	IR signal receiving window, receiving the fixed frequency 38KHz IR remote signal.
4	INPUT / OUTPUT (1~8) buttons	Input/output channel switching buttons. Firstly press the OUTPUT (1-8) button to select the output channel, and the corresponding output channel will flash on the LCD screen. Then press the INPUT (1-8) button to select the input signal source and the corresponding input channel will be displayed on the LCD screen.

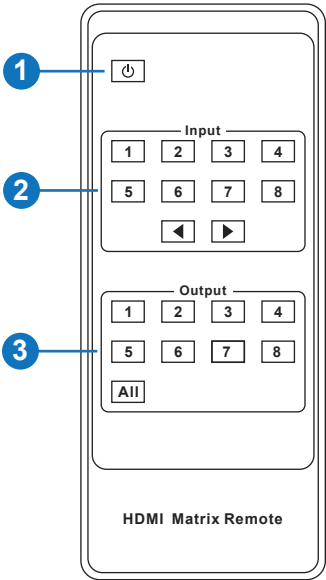
No.	Name	Function Description
5	MENU/ENTER/ UP/DOWN button	MENU: The menu button, with three primary menus (OUTPUT/INPUT/SETUP), which can set various functions of the matrix, including output resolution/color space conversion/mode/signal format/stream, input EDID, buzzer switch, screen display mode switch, baud rate, DHCP switch, restart and reset. UP: Press the UP button to select the previous item. DOWN: Press the DOWN button to select the next item. ENTER: Confirm button. For example: Switch the resolution of output channel 2 to resolution 1080P 60Hz. On the condition that the LCD screen is in default display state and the front panel buttons are not locked, Step 1: Press MENU button, LCD screen will display "MENU->OUTPUT". Step 2: Press ENTER button, LCD screen will display "OUTPUT->RESO". Step 3: Press ENTER button, LCD screen will display two lines of content. The top line is "OUTPUT ->RESO", and the bottom line is "OUT1: 4K * 2K30". Step 4: Press UP/DOWN button to switch to output channel 2. Step 5: Press ENTER button, the current resolution of output channel 2 will flash on the LCD screen. Step 6: Press UP/DOWN button to switch to 1080P60Hz. Step 7: Press ENTER button to confirm, LCD screen will display "SUCCESS", and the switching is completed. Notes: • Pressing the MENU button will return to the previous menu. • In any level menu, it will return to the initial display screen if no operation goes on within 10 seconds.
6	POWER button	Press and hold the button for 3 seconds to enter the standby mode, then short press it to wake up the device.
7	LOCK button	Short press the LOCK button to lock front panel buttons (except the power button); Press it again to unlock.

5.2 Rear Panel



No.	Name	Function Description
1	INPUT 1~8 ports	HDMI signal input ports, connected to HDMI source devices such as PC, DVD or Set-top box with an HDMI cable.
2	AUDIO OUT 1 port	Analog audio output port, only following the audio output of the OUTPUT 1 port.
3	OUTPUT 1~8 ports	HDMI signal output ports, connected to HDMI display devices such as TV or Monitor with an HDMI cable.
4	TCP/IP port	LAN port, used for network input control. Users can access to the Web GUI or Host Computer through this port.
5	RS-232 port	RS-232 serial port, connected to the PC or control system to control the matrix through RS-232 commands transmission or Host Computer operation.
6	DC 12V	DC 12V/2.5A power input port.

6. IR Remote



- ① **Power on or Standby:** Power on the Matrix or set it to standby mode.
 - ② **Input 1/2/3/4/5/6/7/8:** Select the input signal source.
◀ ▶ : Select the previous or next input signal source.
 - ③ **Output 1/2/3/4/5/6/7/8:** Select the output channel.
All: Select all output channels simultaneously. For example, when you press the “All” button and then press input “1” button, at this time the input “1” source will output to all display devices.
- Operation instruction:** You need to press the output button firstly and then press input button to select the corresponding input source. For example,
Press Output-X
(X means output button from 1 to 8 , including “All” button)
Then press Input-Y
(Y means input button from 1 to 8)

7. LCD Screen Menu Navigation

Various functions of the matrix can be set through the operation of buttons on the front panel, and the corresponding content can be displayed on the LCD screen.

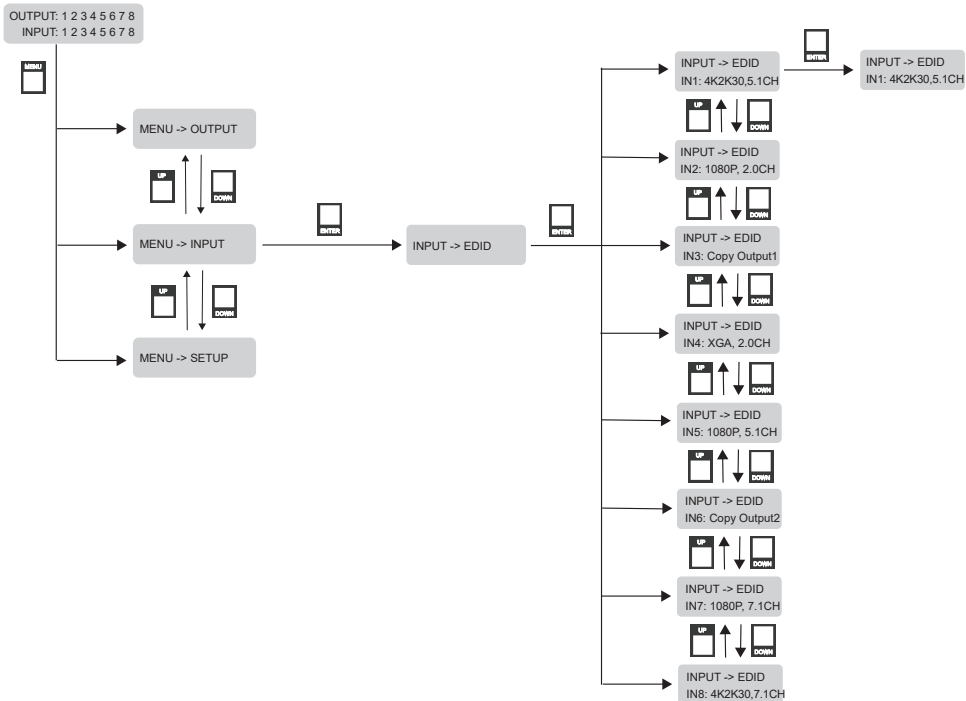
The LCD screen menu navigation consists of three primary menus "OUTPUT/INPUT/SETUP", as well as corresponding secondary and tertiary menus. The specific contents are as follows.

Level 1	Level 2	Level 3
OUTPUT	RESO	4096x2160p 30Hz, 3840x2160p 30Hz, 1920x1200p 60Hz, 1920x1080p 60Hz (default), 1920x1080p 50Hz, 1600x1200p 60Hz, 1600x900p 60Hz, 1440x900p 60Hz, 1360x768p 60Hz, 1280x1024p 60Hz, 1280x720p 60Hz, 1280x720p 50Hz, 1024x768p 60Hz
	CSC	RGB 4:4:4, YCbCr 4:4:4, YCbCr 4:2:2
	MODE	Seamless, Bypass
	FORMAT	HDMI, DVI
	STREAM	Enable, Disable
INPUT	EDID	4K2K30, 2.0CH 4K2K30, 5.1CH 4K2K30, 7.1CH 1080P, 2.0CH 1080P, 5.1CH 1080P, 7.1CH WUXGA, 2.0CH 768P, 2.0CH XGA, 2.0CH User Define1 User Define2 Copy Output1 Copy Output2 Copy Output3 Copy Output4 Copy Output5 Copy Output6 Copy Output7 Copy Output8
SETUP	BEEP	OFF, ON
	LCD ONTIME	OFF, ALWAYS ON, 15 SECOND, 30 SECOND, 60 SECOND
	BAUDRATE	4800, 9600, 19200, 38400, 57600, 115200
	IP INFO	OFF, ON
	REBOOT	/
	RESET	/

8. EDID Management

This matrix has 9 factory defined EDID settings, 2 user-defined EDID modes and 8 copy EDID modes. You can select defined EDID mode or copy EDID mode to input port through front panel buttons, RS-232 command or Web GUI operation.

Front panel buttons operation: On the initial LCD display screen, press “MENU” button to enter the level 1 menu, press “UP/DOWN” button to select INPUT, and then press “ENTER” button. Now the EDID item appears. Press the “ENTER” button, and then press “UP/DOWN” button to select the input channel. Press the “ENTER” button, and then press “UP/DOWN” button to select the EDID mode you need. Finally, press “ENTER” button to confirm, then the LCD screen will prompt “SUCCESS”.



Note: Pressing the “MENU” button will return to the previous menu.

RS-232 command operation: Connect the matrix to the PC with a serial cable, then open a serial command tool on PC to send ASCII command “s edid in x from z!” to set EDID. For details, please refer to “EDID Setting” in the ASCII command list of “10. RS-232 Control Commands”.

Web GUI Operation: Please check the EDID management in the “Input page” of “10. Web GUI User Guide”.

HDMI

Status

Input

Output

Video

System

8x8 HDMI Seamless Matrix Switcher

Admin | Logout | Power On

Input Setting

Inputs	Active	Name	EDID
HDMI 1	☐	Input 1	4K2K30 444, Stereo Audio 2.0
HDMI 2	☐	Input 2	4K2K30 444, Stereo Audio 2.0
HDMI 3	☐	Input 3	4K2K30 444, Stereo Audio 2.0
HDMI 4	☐	Input 4	4K2K30 444, Stereo Audio 2.0
HDMI 5	☐	Input 5	4K2K30 444, Stereo Audio 2.0
HDMI 6	☐	Input 6	4K2K30 444, Stereo Audio 2.0
HDMI 7	☐	Input 7	4K2K30 444, Stereo Audio 2.0
HDMI 8	☐	Input 8	4K2K30 444, Stereo Audio 2.0
All Inputs			4K2K30 444, Stereo Audio 2.0

Load EDID to user memory

Select EDID File:

Browse

Select Destination:

User Define1

Upload

Download EDID to your computer

Select EDID File:

HDMI IN 1

Download

The defined EDID setting list of the product is shown as below:

EDID Mode	EDID Description
1	4K2K30 444, Stereo Audio 2.0
2	4K2K30 444, Dolby/DTS 5.1
3	4K2K30 444, HD Audio 7.1
4	1080P Stereo Audio 2.0
5	1080P Dolby/DTS 5.1
6	1080P HD Audio 7.1
7	1920x1200, Stereo Audio 2.0
8	1360x768, Stereo Audio 2.0
9	1024x768, Stereo Audio 2.0
10	User define1
11	User define2
12	Copy from HDMI output 1
13	Copy from HDMI output 2
14	Copy from HDMI output 3
15	Copy from HDMI output 4
16	Copy from HDMI output 5
17	Copy from HDMI output 6
18	Copy from HDMI output 7
19	Copy from HDMI output 8

Note: When copying a certain output EDID, but this output channel is not connected to a TV, please use the default 1080p, Stereo Audio 2.0 instead.

9. Web GUI Operation Guide

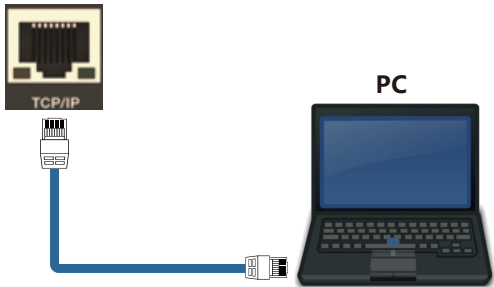
9.1 Connection

The matrix can be controlled by Web GUI. The operation method is shown as below:

Step 1: Get the IP address of the matrix.

Before connecting and logging into the Web GUI, it is necessary to obtain the IP address of the matrix. The default IP address is 192.168.1.100. You can get the current IP address via RS-232 command. Send the ASCII command “ r ip addr!” through a serial command tool, then you'll get the current IP address of the matrix.

Step 2: Connect the TCP/IP port of the matrix to the PC with an UTP network cable, and set the IP address of the PC to be in the same network segment with the matrix.



9.2 Login

Input the current IP address of matrix into the browser on the PC to enter the Web GUI login page.



Select the username and the desired language, then enter the password. The default passwords are as follows.

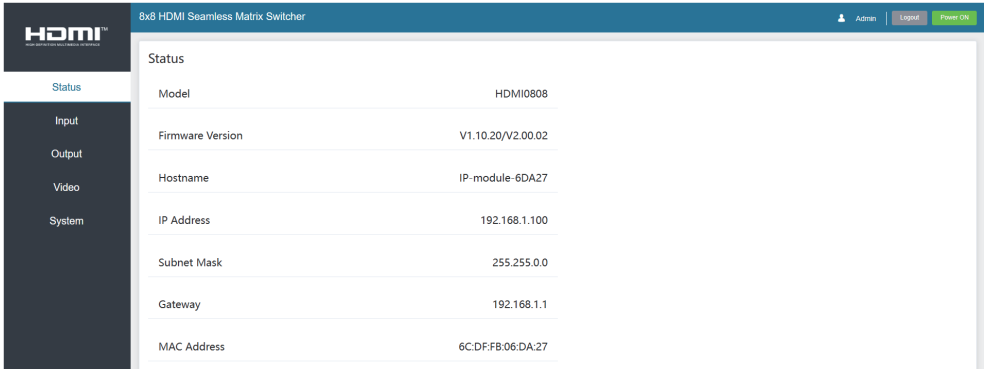
Username	Password
Admin	admin
User	user

After entering the password, click the “LOGIN” button to access to the Web GUI main page.

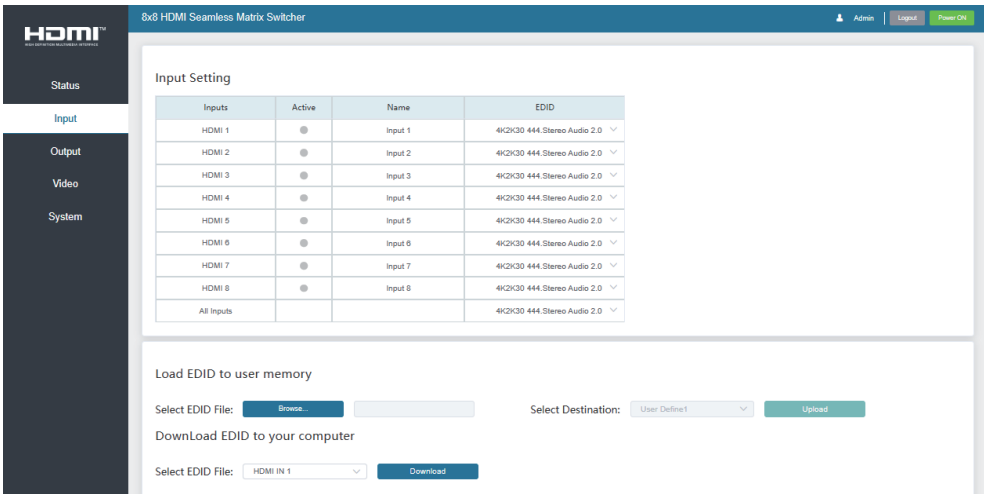
9.3 Web GUI Interface Instruction

■ Status

The Status page provides basic information about the matrix, such as model, firmware version, hostname and the network settings.



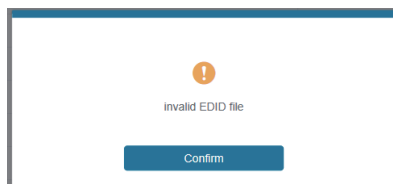
■ Input



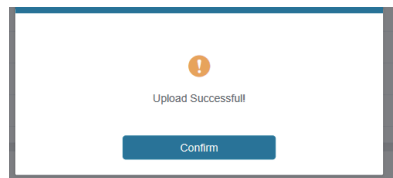
You can do the following operations on the Input page:

- ① **Inputs:** Input channel of the matrix.
- All Inputs:** The settings of All Inputs is available to all inputs above if you select a value from the drop-down list.
- ② **Active:** It indicates whether the channel is connected to a signal source. It is green if the input signal is detected, and gray if no signal.
- ③ **Name:** The input channel's name, which can be modified by entering the corresponding name in the input box. The maximum length of the name is 10 English characters or 5 Chinese characters (numbers and special characters are supported).
- ④ **EDID:** You can set the current channel's EDID. Click drop-down list to select.
- ⑤ **Load EDID to user memory:** Set EDID for the User.

Click the “Browse” button, then select the bin file. If you select the wrong EDID file, there will be a prompt, as shown in the following figure:



Make sure to select the correct file, then you can check the name of the selected file. Select “User Define1” or “User Define2”, then click “Upload”. After successful setting, it will prompt as follows:



⑥ Download EDID to your computer:

Click the drop-down box of “Select EDID File” to select the corresponding input channel. Then click “Download” to download the corresponding EDID file.

■ Output

8x8 HDMI Seamless Matrix Switcher

Admin | Logout | Power On

Status
Input
Output
Video
System

Output Setting

Outputs	Cable	Name	Output Resolution	Color Space	Output Format	Brightness	Contrast Ratio	Stream
HDMI 1		Output 1	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 2		Output 2	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 3		Output 3	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 4		Output 4	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 5		Output 5	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 6		Output 6	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 7		Output 7	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
HDMI 8		Output 8	4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON
All Outputs			4096x2160p30Hz	RGB 4:4:4	HDMI	50	50	OFF ON

You can do the following operations on the Output page:

① **Outputs:** Output channel of the matrix.

All Outputs: The settings of All Outputs is available to all outputs above if you select a value from the drop-down list.

② **Cable:** It indicates the connection status of output ports. When the output port is connected to the display, it shows green, otherwise, it shows gray.

③ **Name:** The output channel's name, which can be modified by entering the corresponding name in the input box. The maximum length of the name is 10 English characters or 5 Chinese characters (numbers and special characters are supported).

④ **Output Resolution:** Set the current output resolution. Click the drop-down list to select other resolutions.

⑤ **Color Space:** Set the color space of the output signal.

⑥ **Output Format:** Set the output signal format (HDMI/DVI).

⑦ **Brightness:** Adjust the brightness of the output signal.

⑧ **Contrast Ratio:** Adjust the contrast ratio of the output signal.

⑨ **Stream:** Turn on/off the signal output stream of the output port.

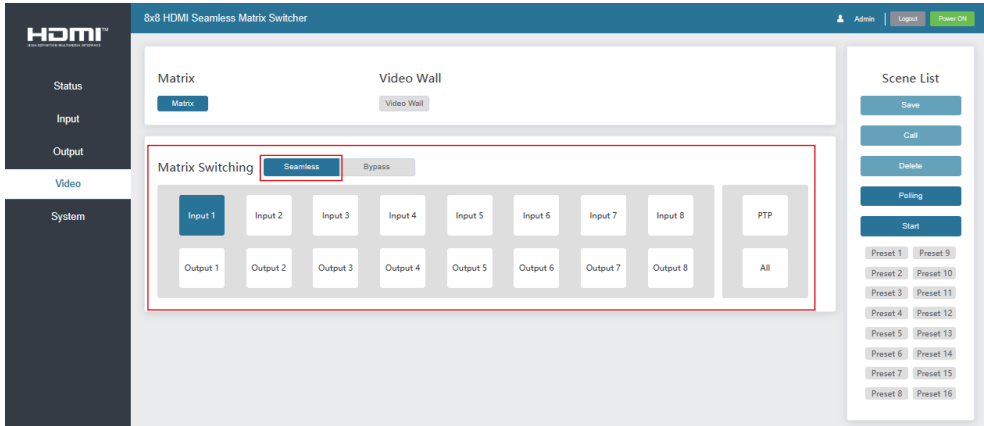
Note: User cannot set each output resolution separately in video wall mode.

■ Video

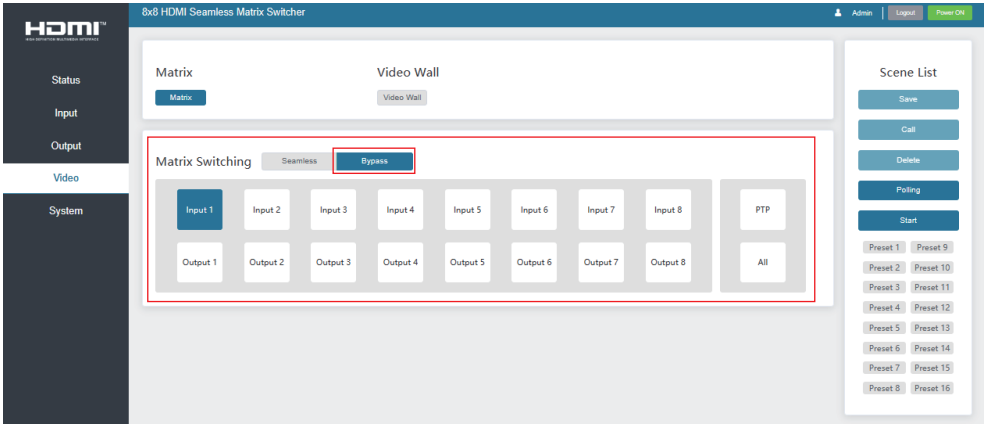
On the video page, you can set two video display modes: Matrix and Video Wall. In each mode, you can set, save, call or delete preset scenes as required, supporting up to 16 presets. In addition, preset scene polling is supported.

- ① **Matrix:** Click “Matrix” to enter the matrix setting mode, then click the corresponding button to select the matrix switching mode (Seamless/Bypass).

Seamless switching mode

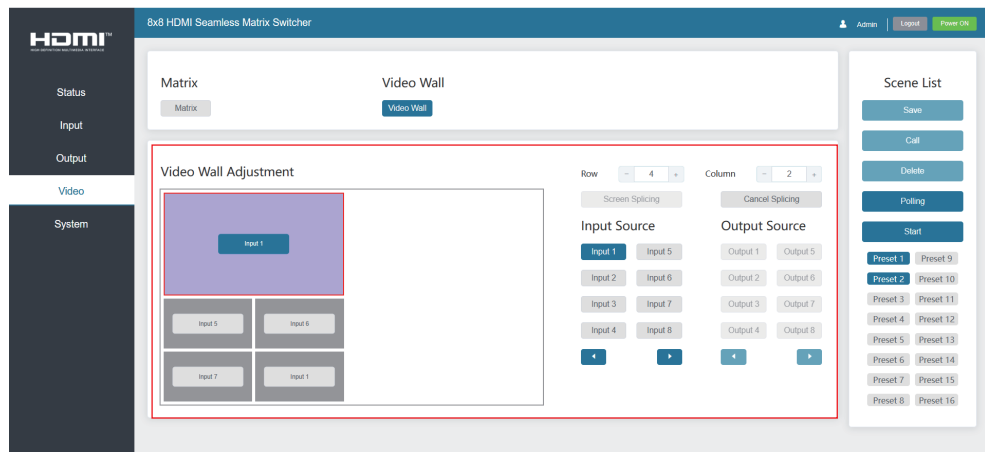


Bypass output mode



Click to select the input source in the Matrix Switching area, then select the output channel that needs to be switched to complete the “one-to-one” or “one-to-many” switching. If you select the input source and then click the “All” button, the “one-to-all” switching is completed, that is, the input signal source will switch to all output channels at the same time. If “PTP” is selected, all input channels and output channels are switched one by one. If switching from non-matrix mode to matrix mode, the corresponding relationship of the original matrix mode will be restored.

② **Video Wall:** Click “Video Wall” to enter the video wall splicing mode.



Follow the steps below to create or configure a video wall.

Step 1: Click “+/-” to set the number of row and column for the video wall.

Step 2: Set the mapping relationship between the splicing screen and the output. The default setting is one-to-one mapping, and users can also click on **Output 1/2/3/4/5/6/7/8** button or ◀/▶ in “Output Source” to select the output channel to be switched.

Step 3: Click on **Input 1/2/3/4/5/6/7/8** button or ◀/▶ in “Input Source” to select the input source.

Step 4: Hold down the left mouse button in “Video Wall Adjustment” and drag to select the screens that need to be spliced.

Step 5: Click the **Screen Splicing** button to complete the video wall splicing settings.

③ **Preset:** The Matrix supports 16 scene presets by default, and users can set, save, call, delete scenes as required, or set preset scene polling. The scene name can be modified, with a maximum length of 10 English characters or 5 Chinese characters (supporting numbers and special characters).

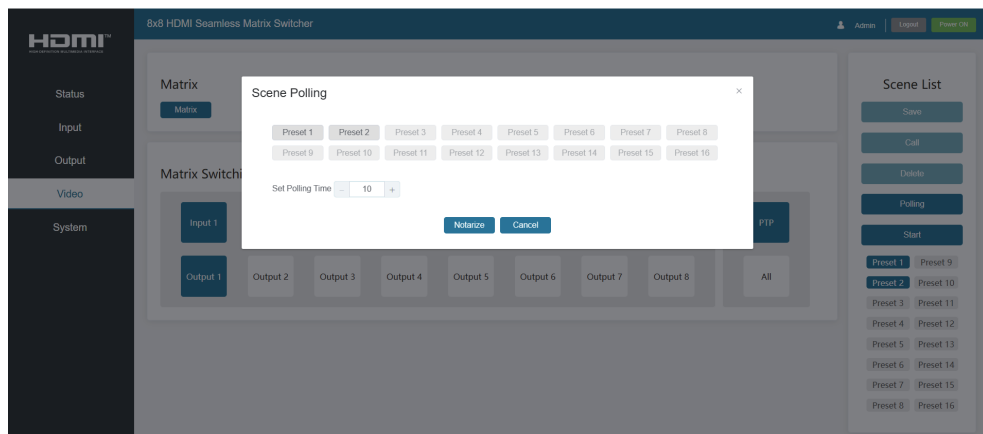
(1) **Preset saving:** After completing the Matrix/Video Wall settings, click the mouse in “Scene List” to select the scene number to be saved, and then click the “Save” button to save the preset scene.

(2) **Preset calling:** Click to select the preset scene name in “Scene List”, then click the “Call” button to call the scene.

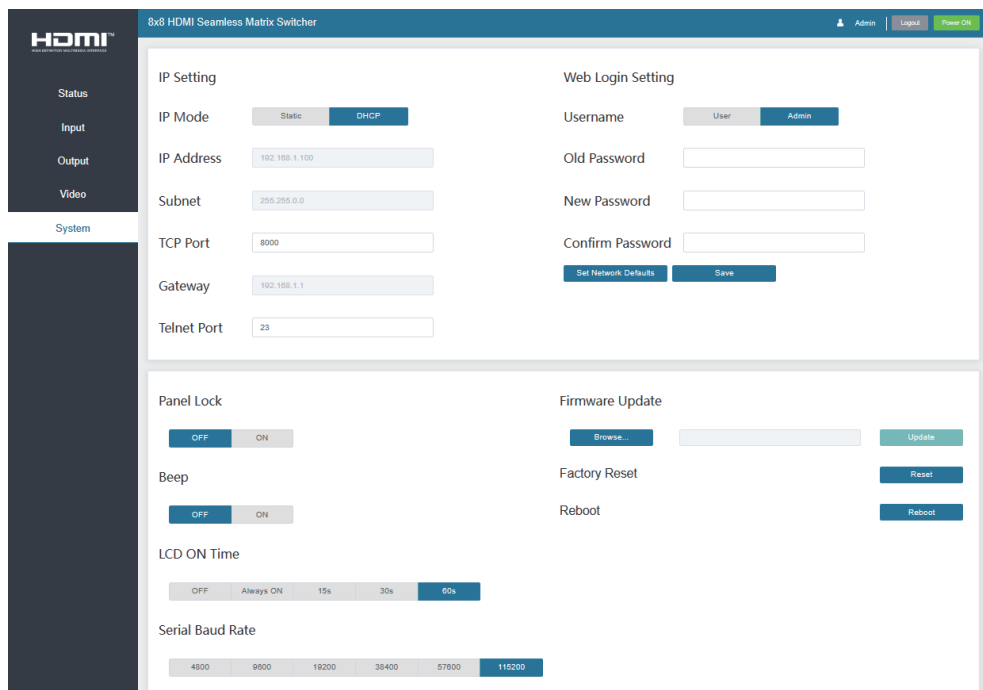
(3) **Preset deleting:** Click to select the preset scene name in “Scene List”, then click the “Delete” button to delete the scene.

(4) **Preset polling:** Click the “Polling” button in “Scene List”, then click to select the saved preset scenes (**Note:** The unsaved scenes are gray, and can not be used for preset polling) in the pop-up Scene Polling window. Click “+/-” to set the polling time (up to 180 seconds), then click “Notarize” to complete the setting. Finally, click the “Start” button in “Scene List” to start preset polling.

Note: Any setting operation (such as switching matrix correspondence relationships) by the users may interrupt the preset polling.



■ System



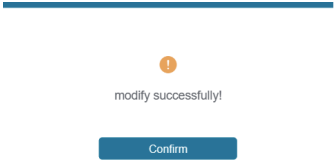
You can do the following operations on the System page:

① Modify Network Setting

You can set the IP Mode (Static/DHCP). If the mode is set to “Static”, you can manually set the IP Address, Gateway, Subnet, Tenet Port and TCP Port as required, then click “Save” to take effect. If the mode is set to “DHCP”, the system will search and fill the IP Address with the one assigned by the router automatically. You can't modify it now.

② Modify Web Login Password

You can modify the Web login password for the User/Admin account. Enter the correct Old Password, New Password, and Confirm Password, then click “Save”. After successful modification, there will be a prompt, as shown in the following figure:

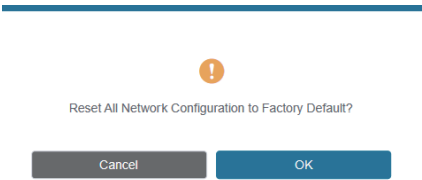


Note: Input rules for changing passwords:

- (1) The password can't be empty.
- (2) New Password can't be the same as Old Password.
- (3) New Password and Confirm Password must be the same.

③ Set the Default Network

Click “Set Network Defaults” button, there will be a prompt, as shown in the following figure:



Click “OK” to search the IP Address again. After searching is completed, it will switch to the login page, the default network setting is completed.

- ④ **Panel Lock:** Click to lock/unlock panel buttons. “ON” indicates that panel buttons are unavailable; “OFF” indicates that panel buttons are available.
- ⑤ **Beep:** Click to turn on/off the beep.
- ⑥ **LCD ON Time:** Click to turn on/off the LCD, or set the LCD display duration time (Always ON/15s/30s/60s).
- ⑦ **Serial Baud Rate:** Click the value to set the Serial Baud Rate.
- ⑧ **Firmware Update:** Click “Browse” to select the update file, then click “Update” to complete firmware update.
- ⑨ **Factory Reset:** Reset the machine to factory defaults by clicking “Reset”.
- ⑩ **Reboot:** Reboot the machine by clicking “Reboot”.

Note: After reset/reboot, it will switch to the login page.

10. RS-232 Control Commands

The product also supports RS-232 command control. Connect the RS-232 port of the product to a PC with an USB to RS-232 serial cable. Then open a Serial Command tool on PC to send ASCII commands to control the product.

The ASCII commands list about the product is shown as below.

ASCII Command

Serial port protocol: baud rate: 115200 (default) Data bits: 8 Stop bits: 1 Check bit: 0
 TCP/IP protocol port: 8000 Telnet port:23

x,y,z, XXX are parameters Error Code describe: E00 -> unown command
 E01 -> parameter out of range E02 -> get the error edid data
 Version: V1.00.01

Command Code	Function Description	Example	Feedback	Default Setting
Power setting				
s power z!	Power on/off the device, z=0~1 (z=0 power off, z=1 power on)	s power 0! s power 1!	power off power on system initializing... initialization finished! fw version 1.00.01	power on
r power!	Get current power state	r power!	power on /power off	
s reboot!	Reboot the device	s reboot!	reboot... system initializing... initialization finished! fw version 1.00.01	
System setting				
help!	Lists all commands	help!		
?!	Lists all commands	?!		
r type!	Get device model	r type!	device type: HDP-MX988SL	
s type *****!	Set device model name	s type HDP-MX988SL!	device type: HDP-MX988SL	
r status!	Get device current status	r status!	get the unit all status: power, beep, lock, in / out connection, video/ audio crosspoint, edid, scaler, network status	
r fw version!	Get firmware version	r fw version!	mcu fw version: v1.00.01 web GUI version: v2.00.01	
r link in x!	Get the connection status of the x input port, x=0~8 (0=all)	r link in 1!	HDMI input 1: connect	
r link out y!	Get the connection status of the y output port, y=0~8 (0=all)	r link out 1!	HDMI output 1: connect	
s reset!	Reset to factory defaults	s reset!	reset to factory defaults system data init system initializing... initialization finished! fw version 1.00.01	
s beep z!	Enable/disable buzzer function, z=0~1(z=0 beep off, z=1 beep on)	s beep 1! s beep 0!	beep on beep off	beep on
r beep!	Get buzzer state	r beep!	beep on / beep off	
s lock z!	Lock/unlock front panel button, z=0~1 (z=0 lock off, z=1 lock on)	s lock 1! s lock 0!	panel button lock on panel button lock off	panel button lock off
r lock!	Get panel button lock state	r lock!	panel button lock on	

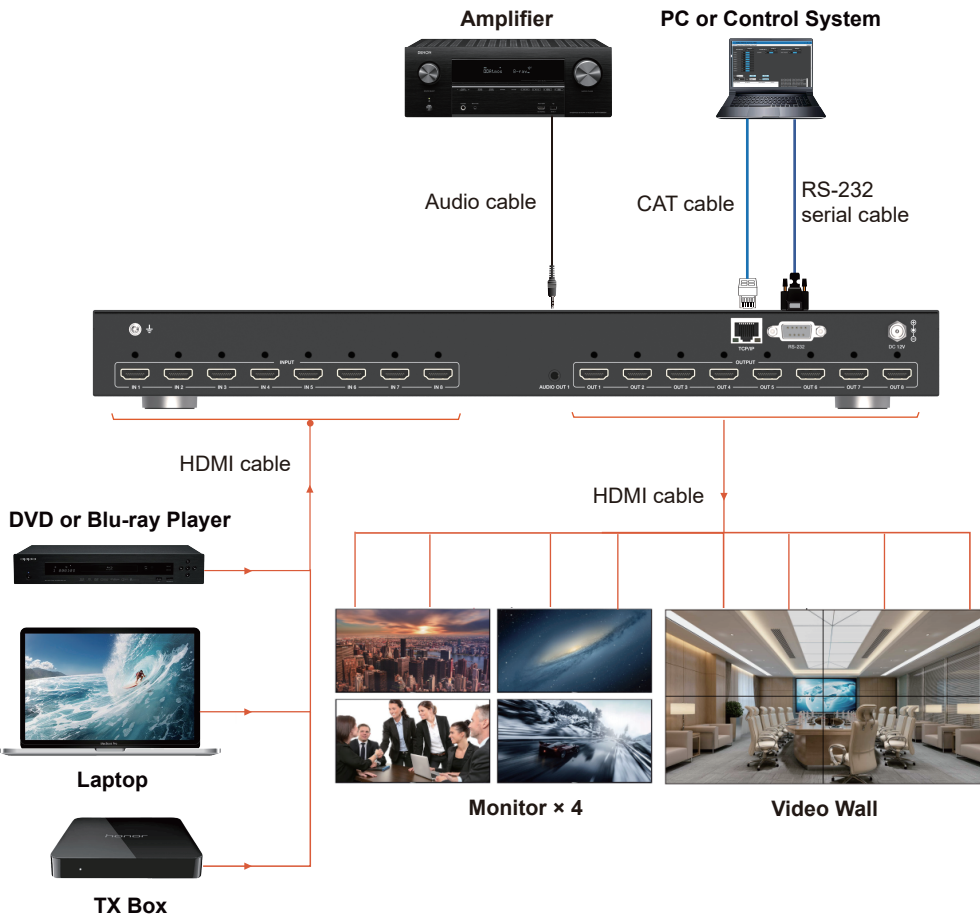
Command Code	Function Description	Example	Feedback	Default Setting
s save preset z!	Save switch state between all output port and the input port to preset z, z=1~16	s save preset 1!	save to preset 1!	
s recall preset z!	Call saved preset z scenarios, z=1~16	s recall preset 1!	recall from preset 1	
s clear preset z!	Clear stored preset z scenarios, z=1~16	s clear preset 1!	clear preset 1!	
r preset z!	Get preset z information, z=1~16	r preset 1!	video/audio crosspoint	
s lcd on time z!	Set LCD screen remain on time, z=0~4 (0:off, 1:always, 2:15s, 3:30s ,4:60s)	s lcd on time 2!	LCD on 15 seconds	LCD on 60 seconds
r lcd mode!	Get the backlight status of LCD screen	r lcd mode!	LCD on 15 seconds	
s baudrate z!	Set baudrate, z=1~6 (1:480, 2:9600, 3:19200, 4:38400, 5:57600, 6:115200)	s baudrate 6!	baudrate :115200	115200
r baudrate!	Get baudrate	r baudrate!	baudrate :115200	
Output setting				
s in x av out y!	Set input x to output y, x=1~8, y=0~8 (0=all)	s in 1 av out 2!	input 1 -> output 2	ptp
r av out y!	Get output y signal status y=0~8 (0=all)	r av out 0!	input 1 -> output 1 input 2 -> output 2 input 3->output 3 input 4 -> output 4 input 5->output 5 input 6->output 6 input 7->output 7 input 8->output 8	
s hdmi y stream z!	Set output y stream on/off, y=0~8 (0=all) z=0~1 (0:disable, 1:enable)	s hdmi 1 stream 1 !	HDMI output 1 stream: enable	enable
r hdmi y stream!	Get output y stream status, y=0~8 (0=all)	r hdmi 1 stream!	HDMI output 1 stream: enable	
s hdmi y res z!	Set HDMI output y resolution z, y=0~8 (0=all), z=1~13 z=1: 4096x2160p30Hz z=2: 3840x2160p30Hz z=3: 1920x1200p60Hz z=4: 1920x1080p60Hz z=5: 1920x1080p50Hz z=6: 1600x1200p60Hz z=7: 1600x900p60Hz z=8: 1440x900p60Hz z=9: 1360x768p60Hz z=10: 1280x1024p60Hz z=11: 1280x720p60Hz z=12: 1280x720p50Hz z=13: 1024x768p60Hz	s hdmi 1 res 4!	HDMI output 1 resolution 1920x1080P60Hz	
r hdmi y res!	Get HDMI output y resolution, y=0~8 (0=all)	r hdmi 1 res!	HDMI output 1 resolution 1920x1080P60Hz	
s hdmi y csc z!	Set HDMI output y color space z, y=0~8 (0=all), z=1~3 (1:RGB444, 2:YCbCr444, 3:YCbCr422)	s hdmi 1 csc 1!	HDMI output 1 color space RGB444	

Command Code	Function Description	Example	Feedback	Default Setting
r hdmi y csc!	Get HDMI output y color space, y=0~8 (0=all)	r hdmi 1 csc!	HDMI output 1 color space RGB444	
s hdmi y format z!	Set HDMI output y format z, y=0~8 (0=all), z=1~2 (1:HDMI, 2:DIV)	s hdmi 1 format 1!	HDMI output 1 format HDMI	
r hdmi y format!	Get HDMI output y format, y=0~8 (0=all)	r hdmi 1 format!	HDMI output 1 format HDMI	
s hdmi y bri z!	Set HDMI output y brightness z, y=0~8 (0=all), z=0~100	s hdmi 1 bri 0!	HDMI output 1 brightness 0	
r hdmi y bri!	Get HDMI output y brightness, y=0~8 (0=all)	r hdmi 1 bri!	HDMI output 1 brightness 0	
s hdmi y con z!	Set HDMI output y contrast z, y=0~8 (0=all), z=0~100	s hdmi 1 con 0!	HDMI output 1 contrast 0	
r hdmi y con!	Get HDMI output y contrast, y=0~8 (0=all)	r hdmi 1 con!	HDMI output 1 contrast 0	
s display mode x!	Set matrix display mode, x=1~2 (1:matrix, 2:video-wall)	s display mode 1!	display mode matrix	
r display mode!	Get matrix display mode	r display mode!	display mode matrix	
EDID setting				
s edid in x from z!	Set input x edid from default edid z, x=0~8 (0=all), z=1~15 z=1: 4K2K30_444, stereo audio 2.0 z=2: 4K2K30_444, dolby/DTS 5.1 z=3: 4K2K30_444, HD audio 7.1 z=4: 1920x1200, stereo audio 2.0 z=5: 1080P, stereo audio 2.0 z=6: 1080P, dolby/DTS 5.1 z=7: 1080P, HD audio 7.1 z=8: 1360x768, stereo audio 2.0 z=9: 1024x768, stereo audio 2.0 z=10: user1 z=11: user2 z=12: copy from HDMI output 1 z=13: copy from HDMI output 2 z=14: copy from HDMI output 3 z=15: copy from HDMI output 4 z=16: copy from HDMI output 5 z=17: copy from HDMI output 6 z=18: copy from HDMI output 7 z=19: copy from HDMI output 8	s edid in 1 from 1! s edid in 0 from 1!	input 1 edid: 4k30, 2.0ch all input edid: 4K30, 2.0ch	1080p, stereo audio 2.0
r edid in x!	Get input edid mode, x=0~8 (0=all)	r edid in 1!	input 1 edid:1080p, 2.0ch	

Command Code	Function Description	Example	Feedback	Default Setting
r edid data hdmi y!	Get the edid data of the HDMI output y port, y=1~8	r edid data hdmi 1!	HDMI out 1 edid: 00 FF FF FF FF FF FF 00 1D D3 46 ED 1B 27 00 00 01 15 01 03 80 65 39 78 2E 0C 55 A2 56 4C 96 25 1A 50 54 BF EF 80 81 80 8B C0 61 40 45 40 81 40 D1 C0 90 40 95 00 02 3A 80 40 71 38 1B 40 40 72 13 00 35 AD 10 00 00 1E 3E 1C 56 A0 50 00 16 30 30 20 35 00 35 AD 10 00 00 1E 00 00 00 FD 00 32 4B 1E 50 10 00 0A 20 20 20 20 20 20 00 00 00 FC 00 47 2D 44 48 44 0A 20 20 20 20 20 20 20 01 11 02 03 1B 71 47 01 03 84 05 12 13 14 23 0F 57 07 83 4F 00 00 66 03 0C 00 10 00 80 01 1D 80 18 71 1C 16 20 58 2C 25 00 C4 8E 21 00 00 9E 8C 0A D0 8A 20 E0 2D 10 10 3E 96 00 13 8E 21 00 00 18 8F 0A D0 8A 20 E0 2D 10 10 3E 96 00 C4 8E 21 00 00 18 8C 0A D0 8A 20 E0 2D 10 10 3E 96 00 13 8E 21 00 00 18 8C 0A D0 8A 20 E0 2D 10 10 3E 96 00 C4 8E 21 00 00 18 00 00 00 00 00 00 00 00 00 00 B7	
Video Wall setting				
creat vw screen row x col y!	Create video wall screen rows and columns layouts, x=1~8, y=1~8	creat vw screen row 2 col 2!	create vw screen 2x2	4x2
s screen x output y!	Set HDMI output y to screen x, x=1~8, y=1~8	s screen 1 output 1!	HDMI output1-> screen 1	
s vw group z row x col y!	Set video wall group z rows and columns, z=1~4, x=1~8, y=1~8 (x*y<=8)	s vw group 1 row 2 col 2!	vw group 1 row2 col 2	
s vw group z screen a!	Set video wall group z screen number, z=1~4, a=1~8	s vw group 1 screen 1!	vw group 1 screen 1	
s vw group z source x!	Set video wall group z select input source, z=1~4, x=1~8	s vw group 1 source 1!	vw group 1 source 1	
delete vw group z!	Delete video wall group z, z=1~4	delete vw group 1!	delete vw group 1!	
r vw info!	Get current video wall scene information	r vw info!	mosaic id:1 mosaic row:2 mosaic col:2 mosaic src:1 mosaic res: 3840x2160 mosaic screen: 1	
Network setting				
r ipconfig!	Get the current IP configuration	r ipconfig !	IP mode: DHCP IP: 192.168.99.100 subnet mask: 255.255.255.0 gateway: 192.168.99.1 TCP/IP port: 8000 telnet port: 23 MAC: 6C:DF:FB:06:F5:EE (static: 192.168.1.100 255.255.255.0 192.168.1.1)	

Command Code	Function Description	Example	Feedback	Default Setting
r mac addr!	Get network MAC address	r mac addr!	MAC address: 6C:DF:FB:06:F5:EE	
s ip mode z!	Set network IP mode to static IP or DHCP, z=0~1 (z=0 static, z=1 DHCP)	s ip mode 0!	IP mode: static (please use "s net reboot" command or repower device to apply new config!)	
r ip mode!	Get network IP mode	r ip mode!	IP mode: DHCP	
s ip addr xxx.xxx.xxx.xxx!	Set network IP address	s ip addr 192.168.1.100!	DHCP on, device can't config static address, set DHCP off first. IP address: 192.168.1.100 (please use "s net reboot" command or repower device to apply new config!)	
r ip addr!	Get network IP address	r ip addr!	IP address: 192.168.99.100	
s subnet xxx.xxx.xxx.xxx!	Set network subnet mask	s subnet 255.255.255.0!	DHCP on, device can't config static address, set DHCP off first. subnet mask: 255.255.255.0 (please use "s net reboot" command or repower device to apply new config!)	
r subnet!	Get network subnet mask	r subnet!	subnet mask: 255.255.255.0	
s gateway xxx.xxx.xxx.xxx!	Set network gateway	s gateway 192.168.1.1!	DHCP on, device can't config static address, set DHCP off first. gateway: 192.168.1.1 (please use "s net reboot" command or repower device to apply new config!)	
r gateway!	Get network gateway	r gateway!	gateway:192.168.99.1	
s tcp/ip port x!	Set network TCP/IP port (x=1~65535)	s tcp/ip port 8000!	TCP/IP port: 8000	
r tcp/ip port!	Get network TCP/IP port	r tcp/ip port!	TCP/IP port: 8000	
s telnet port x!	Set network telnet port (x=1~65535)	s telnet port 23!	set telnet port:23	
r telnet port!	Get network telnet port	r telnet port!	telnet port:23	
s net reboot!	Reboot network modules	s net reboot!	the network is restarting. please wait! IP mode: DHCP IP: 192.168.99.100 subnet mask: 255.255.255.0 gateway: 192.168.99.1 TCP/IP port: 8000 telnet port: 23 MAC: 6C:DF:BB:06: F5:EE (static: 192.168.1.100 255.255.255.0 192.168.1.1)	

11. Application Example



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